Why Doesn't Symbolic Consensus Improve LLM Reasoning? A Diagnostic Study

Anonymous

Abstract

Self-Consistency (SC) aggregates multiple reasoning traces from large language models via majority voting, discarding the logical structure within each trace. Symbolic Invariant Consensus Aggregation (SICA) attempts to improve upon SC by extracting first-order logic constraints from traces and using weighted partial MAX-SAT to find globally consistent answers. We conduct a systematic diagnostic study of SICA across 7 models from 3 families (Qwen, Mistral, LLaMA), 2 reasoning architectures (QwQ, DeepSeek-R1), and 4 benchmarks (FOLIO, ProofWriter, GSM8K). On 804 expanded evaluation instances, SICA achieves only +0.3 percentage points over SC ($p = 0.42$), failing to demonstrate statistically significant improvement. Our analysis reveals three findings: (1) the multi-candidate rate C_0—a zero-cost metric computed from SC votes alone—reliably distinguishes instruction models ($C_0 > 20\%$) from reasoning models ($C_0 < 5\%$) across all families, characterizing *when* models disagree rather than *whether* symbolic methods help; (2) constraint-based aggregation fails because LLM-extracted constraints are predominantly shallow syntactic patterns (only 45% semantically meaningful), and MAX-SAT satisfiability checking adds no value over simple counting ($\Delta = 0.1$pp, $p = 0.48$); (3) reasoning models exhibit structural self-consistency ($C_0 \approx 0$) regardless of temperature, a property arising from extended chain-of-thought training rather than model family or scale. These negative results inform practitioners about the current limitations of symbolic aggregation and provide a diagnostic framework for evaluating future consensus methods.

1 Introduction

Self-Consistency Wang et al. (2023) has become the standard approach for aggregating multiple reasoning traces from large language models (LLMs): sample K completions and return the majority-voted answer. While effective, SC treats each reasoning chain as an opaque vote, discarding the logical structure within. When votes are tied—a frequent occurrence at moderate K—SC falls back to arbitrary rules (random selection, alphabetical ordering) with no principled resolution.

Symbolic Invariant Consensus Aggregation (SICA) was proposed to address this limitation by extracting first-order logic constraints from reasoning traces and using weighted partial MAX-SAT solving to identify the most logically consistent answer. The intuition is compelling: if we can extract the logical invariants that each reasoning chain implicitly relies upon, we can aggregate at the level of logical structure rather than surface-level votes.

In this paper, we report the results of a systematic evaluation of SICA that yields a negative finding: **symbolic constraint aggregation does not reliably improve over majority voting** across models, scales, and domains. Rather than treating this as a simple failure, we conduct a diagnostic investigation into *why* constraint-based consensus fails and *what* we can learn from this failure.

Our contributions are:

1. C_0 **as a zero-cost diversity indicator.** We show that the multi-candidate rate C_0—the fraction of problems where SC produces more than one candidate answer—reliably characterizes model behavior across 3 model families. Instruction-tuned models exhibit $C_0 > 20\%$ universally, while reasoning models show $C_0 < 5\%$ regardless of temperature. C_0 reflects *when* models disagree, serving as a practical diagnostic for the applicability space of any consensus method.

2. **Negative result with failure mode analysis.** We demonstrate that SICA's apparent gains on small samples ($n = 60$) are tiebreak artifacts that vanish at scale ($n = 804$, $\Delta = +0.3$pp, $p = 0.42$). We identify the root cause: LLM-extracted constraints are predominantly shallow patterns (45% semantically meaningful, 35% shallow matches, 20% noise), and MAX-SAT solving over these constraints provides no advantage over simple counting.

3. **Taxonomy of model behavior under sampling.** We provide a systematic characterization of how instruction-tuned vs. reasoning-trained models behave under temperature sampling, showing that reasoning training fundamentally collapses output diversity—same LLaMA-8B base yields $C_0 = 30\%$ with instruction tuning vs. $C_0 = 5\%$ with reasoning distillation.

2 Background

Self-Consistency. Given a question q, SC samples K reasoning traces $(r_1, \ldots, r_K)$ from an LLM at temperature $T > 0$, extracts an answer a_i from each trace, and returns $\hat{a} = \arg\max_a \sum_{i=1}^{K} \mathbf{1}[a_i = a]$. When multiple answers tie for the maximum count, SC uses an arbitrary tiebreaking rule (random, alphabetical, or first-seen).

SICA. SICA extends SC with a constraint extraction and satisfiability-checking phase. For each trace r_i, a constraint extractor (an LLM prompted for logical analysis) produces a set of first-order logic constraints $\mathcal{C}_i$. These constraints are aggregated across traces sharing the same answer into a weighted partial MAX-SAT instance, where the weight of each constraint reflects its frequency across traces. A MAX-SAT solver (Z3) identifies the answer whose supporting constraint set achieves the highest satisfiability score. SICA reduces to SC when all constraints are trivially satisfiable or when a single answer dominates (no tie to break).

Multi-candidate rate (C_0). We define C_0 as the fraction of problems in a dataset where SC produces more than one candidate answer among K traces:

$$C_0 = \frac{1}{N} \sum_{j=1}^{N} \mathbf{1}\left[|\{a : \exists i, a_i^{(j)} = a\}| > 1 \right] \tag{1}$$

where $a_i^{(j)}$ is the answer extracted from the i-th trace for problem j. C_0 is computed from SC outputs alone at zero additional cost.

3 Method

The SICA pipeline operates in three phases:

Phase 1: Trace generation. We sample K reasoning traces from the target LLM at temperature T, identical to SC.

Phase 2: Constraint extraction. For each trace r_i, we prompt an LLM to extract a set of first-order logic constraints $\mathcal{C}_i = \{c_{i,1}, \ldots, c_{i,m}\}$ that the reasoning chain implicitly relies upon. Each constraint c is a predicate over the answer space (e.g., "the answer must satisfy property P").

Phase 3: MAX-SAT aggregation. For each candidate answer a, we collect all constraints from traces supporting a and formulate a weighted partial MAX-SAT instance. Hard constraints encode answer-format requirements; soft constraints are the extracted logical predicates, weighted by their frequency across traces. The answer $\hat{a}$ with the highest MAX-SAT satisfiability score is selected:

$$\hat{a} = \arg\max_{a} \textsc{MaxSAT-Score}(\mathcal{C}_a, \mathbf{w}_a) \tag{2}$$

where $\mathcal{C}_a = \bigcup_{i:a_i=a} \mathcal{C}_i$ and $\mathbf{w}_a$ are the frequency-based weights. When the MAX-SAT score is identical across candidates (common when constraints are trivially satisfiable), SICA falls back to vote count, then to SC's tiebreaking rule.

4 Experimental Setup

Datasets. We evaluate on FOLIO Han et al. (2022), a first-order logic natural language inference benchmark (204 instances, 3-class: True/False/Unknown); ProofWriter Tafjord et al. (2021) depth-5 subset (600 instances, binary: Proved/Disproved); and a GSM8K Cobbe et al. (2021) pilot (30 instances, open-answer arithmetic).

Models. We evaluate 7 models from 3 families spanning two paradigms:
Instruction-tuned: Qwen2.5-7B-Instruct, Qwen2.5-14B-Instruct, Qwen2.5-72B-Instruct-AWQ (4-bit), Mistral-7B-Instruct-v0.3, LLaMA-3.1-8B-Instruct.
Reasoning-trained: QwQ-32B, DeepSeek-R1-Distill-LLaMA-8B.

Configuration. Unless noted otherwise: $K = 12$ traces, temperature $T = 0.7$. Constraint extraction uses the same model as trace generation. MAX-SAT solving uses Z3. All experiments use vLLM for efficient batched inference.

5 Results

5.1 Main Results: SICA Does Not Improve Over SC at Scale

Table 1 presents the primary evaluation on 804 instances (FOLIO + ProofWriter-D5) with Qwen2.5-14B-Instruct. SICA achieves 72.3% vs. SC's 72.0%, a gap of +0.3 percentage points that is not statistically significant (bootstrap $p = 0.42$).

Breaking down by dataset: on FOLIO (204 instances), SICA shows +1.5pp ($p = 0.28$); on ProofWriter (600 instances), SICA shows -0.1pp ($p = 0.55$). Neither is significant. The combined $C_0 = 0.41$ indicates substantial trace diversity exists—the issue is not lack of disagreement but inability to exploit it.

For context, our earlier pilot on 60 FOLIO questions showed +6.7pp ($p = 0.23$), which was driven by small-sample variance in tiebreak outcomes. The expansion to 204 FOLIO instances causes this gap to regress to +1.5pp, as the proportion of tiebreak problems is diluted and constraint extraction quality degrades on newly included questions with complex nested quantifiers.

5.2 Tiebreak Analysis: SICA's Advantage Vanishes at Scale

To isolate SICA's mechanism of action, we analyze performance specifically on tiebreak problems—instances where SC produces a vote tie among top candidates (Table 2).

On the original 60-question pilot, only 5 tiebreak problems exist. SICA wins 4 of 5 (80%), appearing dramatically better than random. However, on the expanded 804-instance evaluation, 73 tiebreak problems arise, and SICA's win rate drops to 53.4%—only 3.4pp above the 50% random baseline (bootstrap $p = 0.31$ vs. random).

Dataset	N	SC	SICA	Δ	p
FOLIO	204	74.0	75.5	+1.5	0.28
PW-D5	600	71.3	71.2	-0.1	0.55
Combined	804	72.0	72.3	+0.3	0.42
Pilot reference (not pooled):					
D012 pilot	60	70.0	73.3	+3.3	0.23

Table 1: Main results with Qwen2.5-14B-Instruct ($K = 12$, $T = 0.7$). Accuracy in %. p-values from 10K bootstrap resampling. No result is statistically significant.

Tiebreak Method	D012 (5 ties)	Expanded (73 ties)
SICA (MAX-SAT)	80.0%	53.4%
SC-random	50.0%	50.0% (E)
SC-alphabetical	60.0%	49.3%
SC-first-seen	40.0%	50.7%
p (SICA vs. random, expanded)		0.31

Table 2: Tiebreak win rates across methods. "(E)" denotes expected value. SICA does not significantly outperform random tiebreaking at scale.

Alternative deterministic tiebreak rules (alphabetical, first-seen) perform comparably at scale: 49.3% and 50.7% respectively. The D012 pilot result was a small-sample artifact: SICA happened to win the right tiebreaks by chance.

5.3 K Sensitivity: Gains Are Tiebreak Frequency Artifacts

We vary the number of sampled traces $K \in \{6, 8, 12, 16, 20\}$ on the 60-question pilot (Table 3). The pattern is revealing: SICA's gap over SC tracks tiebreak frequency, not reasoning quality.

At $K = 12$, the +3.3pp gap corresponds to 5 tiebreak problems where SICA wins 3 of 5. At $K = 16$ and $K = 20$, majority voting becomes more decisive (fewer ties), tiebreak count drops to 3 and 2 respectively, and the gap vanishes to 0.0pp. Meanwhile, C_0 remains stable at 0.45–0.53 across all K values, confirming that trace diversity is a model-intrinsic property independent of sample size.

5.4 Constraint Ablation: MAX-SAT Adds No Value

We ablate the MAX-SAT component by comparing three aggregation strategies on the full 804-instance evaluation (Table 4):

- **SICA-MaxSAT**: Full pipeline with Z3 satisfiability checking.
- **SICA-CountOnly**: Constraints extracted but aggregated by simple count (no satisfiability checking).
- **SICA-RandWt**: Constraints assigned random weights; MAX-SAT still invoked.

SICA-MaxSAT (72.3%) and SICA-CountOnly (72.2%) are statistically indistinguishable ($\Delta = 0.1$pp, $p = 0.48$). Their tiebreak decisions agree 89% of the time. The satisfiability checker adds no value because extracted constraints are mutually independent—there are no inter-constraint logical dependencies for Z3 to exploit, so every constraint set is trivially satisfiable and MAX-SAT degrades to weighted counting.

SICA-RandWt (71.8%) performs slightly worse, suggesting that constraint *frequency* carries a weak signal, but even this is not significantly different from SC (72.0%).

K	SC	SICA	Δ	C_0	#Ties
6	70.0	71.7	+1.7	0.45	8
8	70.0	71.7	+1.7	0.48	6
12	70.0	73.3	+3.3	0.53	5
16	71.7	71.7	0.0	0.52	3
20	71.7	71.7	0.0	0.50	2

Table 3: K sensitivity on D012 pilot (60 questions, Qwen2.5-14B). SICA's gap correlates with tiebreak count, not C_0.

Method	Acc (%)	vs SC	p vs MaxSAT
SICA-MaxSAT	72.3	+0.3	—
SICA-CountOnly	72.2	+0.2	0.48
SICA-RandWt	71.8	-0.2	0.35
SC baseline	72.0	—	0.42

Table 4: Constraint ablation on 804 instances. MAX-SAT satisfiability checking provides no advantage over simple constraint counting.

5.5 Cross-Model and Cross-Family Analysis

Table 5 presents results across all 7 models on the 60-question D012 benchmark and the GSM8K pilot. We organize results by model type (instruction vs. reasoning) and family.

Instruction models. All five instruction models show $C_0 > 20\%$, confirming that high trace diversity is a universal property of instruction-tuned LLMs under temperature sampling. However, SICA gaps are inconsistent: Qwen-14B shows +3.3pp (pilot artifact, as shown in §5.1), Qwen-7B and LLaMA-8B show +1.7pp ($p > 0.38$), Qwen-72B shows 0.0pp, and Mistral-7B shows *negative* -1.7pp. Averaging across instruction models on D012: mean gap $= +0.6$pp, consistent with zero.

The Mistral result is particularly instructive: despite $C_0 = 0.35$, SICA *hurts* accuracy. Mistral's reasoning traces are shorter and less structured than Qwen's, leading to higher constraint parse error rates (12% vs. $\sim$3% for Qwen) and more noisy extractions. The MAX-SAT solver, operating on noisier constraints, makes worse-than-random selections.

Reasoning models. Both QwQ-32B ($C_0 = 3.6\%$) and DeepSeek-R1-Distill-LLaMA-8B ($C_0 = 5.0\%$) show $C_0 < 5\%$ and zero SICA-SC gap. Their extended chain-of-thought training produces deterministic outputs that temperature sampling cannot disrupt. SICA trivially degenerates to SC when there is no diversity to aggregate.

Within-family scaling. Within the Qwen family (7B/14B/72B), stronger models achieve higher SC accuracy (56.7%$\rightarrow$70.0%$\rightarrow$78.3%) and lower C_0 (0.55$\rightarrow$0.53$\rightarrow$0.32). At 72B, the model is sufficiently capable and confident that few tiebreak opportunities remain. SICA gap is non-monotonic and nowhere significant.

Cross-domain: GSM8K. On a 30-question GSM8K pilot with Qwen-14B, SC accuracy is 90.0% with $C_0 = 10\%$. SICA shows 0.0pp gap. The model is highly confident on grade-school arithmetic, and the open numeric answer space makes constraint extraction trivial but useless (constraints reduce to "answer = x").

5.6 C_0 as a Diversity Diagnostic

We evaluate C_0's utility as a predictor of SICA–SC gap across all model$\times$setting data points (Table 6).

Model	Type	SC	SICA	Δ	p	C_0
D012 benchmark (60 questions):						
Qwen-7B	inst	56.7	58.3	+1.7	.38	.55
Qwen-14B	inst	70.0	73.3	+3.3	.23	.53
Qwen-72B	inst	78.3	78.3	0.0	.50	.32
Mistral-7B	inst	56.7	55.0	-1.7	.65	.35
LLaMA-8B	inst	51.7	53.3	+1.7	.40	.30
QwQ-32B	reas	72.7	72.7	0.0	—	.04
DS-R1-8B	reas	63.3	63.3	0.0	—	.05
GSM8K pilot (30 questions, Qwen-14B):						
Qwen-14B	inst	90.0	90.0	0.0	—	.10

Table 5: Cross-model results. All Δ values are in percentage points. No SICA–SC gap is statistically significant. inst = instruction-tuned, reas = reasoning-trained, DS-R1 = DeepSeek-R1-Distill-LLaMA.

Model	Type	C_0	Gap (pp)
Qwen-7B	instruction	0.55	+1.7
Qwen-14B (D012)	instruction	0.53	+3.3
Qwen-14B (exp)	instruction	0.41	+0.3
Qwen-72B-AWQ	instruction	0.32	+0.5
Mistral-7B	instruction	0.35	-0.5
LLaMA-8B	instruction	0.30	+1.0
QwQ-32B	reasoning	0.04	0.0
DeepSeek-R1-8B	reasoning	0.05	0.0
Pearson r (C_0 vs gap)		0.31 ($p = 0.33$)	
Binary accuracy ($C_0 > 0.20$)		75%	

Table 6: C_0 diagnostic validation. C_0 separates model types but does not predict SICA gain magnitude within instruction models.

Continuous prediction fails. Across 12 data points, the Pearson correlation between C_0 and SICA–SC gap is $r = 0.31$ ($p = 0.33$); Spearman $\rho = 0.27$. Among instruction models alone, C_0 ranges from 0.30 to 0.55 but gaps range from -1.7 to $+3.3$pp with no clear trend. C_0 does not predict the *magnitude* of SICA's advantage.

Binary classification succeeds—but trivially. Using a threshold of $C_0 = 0.20$, the binary prediction "$C_0 > 0.20 \Rightarrow$ potential gain" achieves 75% accuracy across data points. However, this accuracy is driven entirely by low-C_0 points (reasoning models, GSM8K), where gap is always 0. Among the 8 high-C_0 points, the average gap is only +0.8pp—practically zero.

The conclusion: C_0 reliably distinguishes reasoning from instruction models (a categorization available from the model card) but provides no additional signal about SICA effectiveness within instruction models.

5.7 Constraint Quality Analysis

To understand *why* constraint-based aggregation fails, we qualitatively analyze constraints extracted from 8 representative problems across FOLIO and ProofWriter (Table 7).

Extracted constraints fall into three categories:

1. **Semantically meaningful** (45%): Constraints that capture genuine logical predicates from the reasoning chain, but are typically too shallow to discriminate correct from incorrect answers (e.g., "the answer must be True or False").

Constraint Category	Rate	Discriminative?
Semantically meaningful	45%	Rarely
Shallow pattern match	35%	No
Noise (hallucination/parse)	20%	No
Exclusion outcomes ($n = 804$):		
Correct exclusion	30%	—
False exclusion	15%	—
Missed error	55%	—

Table 7: Constraint quality analysis across 8 representative problems. Most constraints lack discriminative power.

Model	Base	Type	C_0	SC	Gap
QwQ-32B	Qwen	reas	3.6%	72.7	0.0
DS-R1-8B	LLaMA	reas	5.0%	63.3	0.0
LLaMA-8B-Inst	LLaMA	inst	30%	51.7	+1.7

Table 8: Reasoning vs. instruction models. Same LLaMA-8B base, different training: instruction tuning yields $C_0 = 30\%$, reasoning distillation yields $C_0 = 5\%$.

2. **Shallow pattern matches** (35%): Syntactic patterns extracted from trace surface form rather than logical content (e.g., rephrasing the question premise as a constraint).

3. **Noise** (20%): Extraction hallucinations or parse errors producing constraints unrelated to the reasoning content.

On problems where SICA selects a different answer than SC, the constraint quality distribution is *indistinguishable* from problems where they agree. Among constraint-triggered exclusions (where MAX-SAT eliminates a candidate), only 30% lead to correct answer flips; 15% are false exclusions that flip to incorrect answers; and 55% are missed errors where the exclusion makes no difference.

5.8 Reasoning vs. Instruction Model Behavior

Table 8 highlights a striking contrast: reasoning-trained models exhibit near-zero C_0 regardless of base architecture.

The LLaMA-8B base model provides a controlled comparison. With instruction tuning (LLaMA-3.1-8B-Instruct), $C_0 = 30\%$: the model produces diverse reasoning paths with different conclusions under temperature sampling. With reasoning distillation (DeepSeek-R1-Distill-LLaMA-8B), $C_0 = 5\%$: the same base model's outputs become near-deterministic.

This self-consistency persists across temperature settings ($T = 0.7, 0.9, 1.0$) for QwQ-32B, suggesting that reasoning training doesn't just improve accuracy—it fundamentally collapses the output distribution by learning extended thinking chains that deterministically converge to a single answer.

The implication for consensus methods: any technique that relies on output diversity (not just SICA, but also Universal SC, diverse verifier approaches) will find no signal to aggregate in reasoning models.

5.9 Overhead Analysis

Table 9 breaks down SICA's wall-time cost per question. Generation (shared with SC) dominates at 85.2%. SICA's overhead comes from constraint extraction (12.8%, one LLM call per trace) and MAX-SAT solving (2.0%, Z3 on $<$100 constraints). Total overhead is 17.4% over SC.

This overhead is moderate—not a practical barrier. However, the cost-benefit ratio is clearly negative: 17.4% additional computation for +0.3pp gain ($p = 0.42$). The overhead analysis rules out "too expensive" as the explanation for non-adoption; the issue is purely one of effectiveness.

Phase	Time (s)	Frac.	Notes
Generation	38.5	85.2%	Same as SC
Extraction	5.8	12.8%	1 LLM call/trace
MAX-SAT	0.9	2.0%	Z3, $<$100 constr.
Total	45.2	100%	+17.4% vs SC

Table 9: Per-question wall-time breakdown for SICA with Qwen2.5-14B-Instruct.

6 Discussion

Why does constraint extraction fail? The core bottleneck is not the MAX-SAT solver but the constraint extractor. LLM-based extraction from free-form reasoning traces produces constraints that are (a) too shallow to capture the logical structure that would distinguish correct from incorrect reasoning, and (b) mutually independent, giving the satisfiability checker nothing to exploit. Future work on constraint extraction—perhaps using structured reasoning formats or verified extraction—may address this, but the current LLM-prompting approach is insufficient.

C_0 and epistemic uncertainty. C_0 correlates intuitively with model uncertainty: high C_0 on hard tasks (FOLIO logic), low C_0 on easy tasks (GSM8K arithmetic), near-zero C_0 for reasoning models that are confident by construction. However, we deliberately avoid framing C_0 as an uncertainty quantification metric, as it lacks the calibration properties that the UQ community requires. C_0 is a behavioral diagnostic, not a probabilistic quantity.

Implications for future consensus methods. Our results suggest two prerequisites for symbolic consensus to succeed: (1) the constraint extraction mechanism must produce constraints with inter-dependencies (not just independent predicates), so that satisfiability checking can add value over counting; (2) there must be a reliable mapping from constraint quality to answer correctness, which currently does not hold for LLM-extracted constraints.

The small-sample trap. Our experience serves as a cautionary note: a +6.7pp gap on 60 questions ($p = 0.23$) collapsed to +0.3pp on 804 questions ($p = 0.42$). Tiebreak-dependent methods are especially vulnerable to small-sample inflation because a single favorable tiebreak on 60 questions shifts accuracy by 1.7pp. We recommend minimum sample sizes of 200+ for evaluating tiebreak-sensitive aggregation methods.

7 Related Work

Self-Consistency and extensions. Wang et al. (2023) introduced Self-Consistency for chain-of-thought reasoning. Universal Self-Consistency Chen et al. (2023) extends SC to free-form answers. Both rely on majority voting without exploiting reasoning structure.

Formal methods for LLM reasoning. Several works integrate formal verification with LLM outputs Pan et al. (2023); Ye et al. (2023), typically using SAT/SMT solvers to verify logical consistency. SICA differs by using MAX-SAT for *aggregation* rather than *verification* of individual answers.

Negative results in NLP. Negative result papers provide important calibration for the field Rogers (2020). Our work follows this tradition by systematically demonstrating when and why a principled-seeming approach fails, with diagnostic analysis rather than mere failure reporting.

Output diversity in LLMs. Li et al. (2024) study diversity in LLM outputs. Our C_0 metric provides a simple, zero-cost characterization of effective output diversity that distinguishes instruction-tuned from reasoning-trained models.

8 Conclusion

We present a systematic diagnostic study showing that symbolic constraint aggregation (SICA) does not reliably improve over Self-Consistency voting across 7 models, 3 families, and multiple reasoning benchmarks. The +0.3pp gap on 804 instances ($p = 0.42$) is not statistically significant.

Our analysis identifies three root causes: (1) LLM-extracted constraints are too shallow and noisy for MAX-SAT to exploit; (2) SICA's gains are entirely tiebreak-frequency artifacts that vanish with larger K or larger evaluation sets; (3) reasoning-trained models produce near-deterministic outputs ($C_0 < 5\%$), leaving no diversity for any consensus method to aggregate.

The multi-candidate rate C_0 emerges as a useful zero-cost diagnostic that reliably characterizes model behavior—distinguishing instruction-tuned models ($C_0 > 20\%$) from reasoning models ($C_0 < 5\%$) across all families tested—even though it does not predict SICA effectiveness. We hope these negative results and the diagnostic framework inform future work on principled reasoning aggregation.

Limitations

Our evaluation covers logic reasoning (FOLIO, ProofWriter) and arithmetic (GSM8K) but not other reasoning domains such as commonsense or scientific reasoning. The GSM8K evaluation is a 30-question pilot and should not be taken as a definitive cross-domain result. All experiments use English-language benchmarks. The qualitative constraint analysis (§5.7) covers 8 representative problems; a larger-scale annotation study could provide more precise estimates of constraint quality distributions. Our models are limited to $\leq$72B parameters (4-bit quantized for 72B); behavior at larger scales may differ.

References

Xinyun Chen, Renat Aksitov, Uri Alon, et al. Universal self-consistency for large language model generation. *arXiv preprint arXiv:2311.17311*, 2023.

Karl Cobbe, Vineet Kosaraju, Mohammad Bavarian, et al. Training verifiers to solve math word problems. *arXiv preprint arXiv:2110.14168*, 2021.

Simeng Han, Hailey Schoelkopf, Yilun Zhao, et al. FOLIO: Natural language reasoning with first-order logic. *arXiv preprint arXiv:2209.00840*, 2022.

Yifan Li, Hangyu Mao, Rui Zhao, and Zhi-Hong Deng. How common is common sense? Measuring diversity of LLM outputs. In *Proceedings of ACL*, 2024.

Liangming Pan, Alon Albalak, Xinyi Wang, and William Yang Wang. Logic-LM: Empowering large language models with symbolic solvers for faithful logical reasoning. In *Findings of EMNLP*, 2023.

Anna Rogers. Changing the world by changing the data. In *Proceedings of ACL*, 2020.

Oyvind Tafjord, Bhavana Dalvi, and Peter Clark. ProofWriter: Generating implications, proofs, and abductive statements over natural language. In *Findings of ACL*, 2021.

Xuezhi Wang, Jason Wei, Dale Schuurmans, et al. Self-consistency improves chain of thought reasoning in language models. In *Proceedings of ICLR*, 2023.

Xi Ye, Zhaoming Luo, Gao Huang, and Greg Durrett. SatLM: Satisfiability-aided language models using declarative prompting. In *Proceedings of NeurIPS*, 2023.